UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 7 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 (5 marks)

Study Figure 1 and answer the following questions.

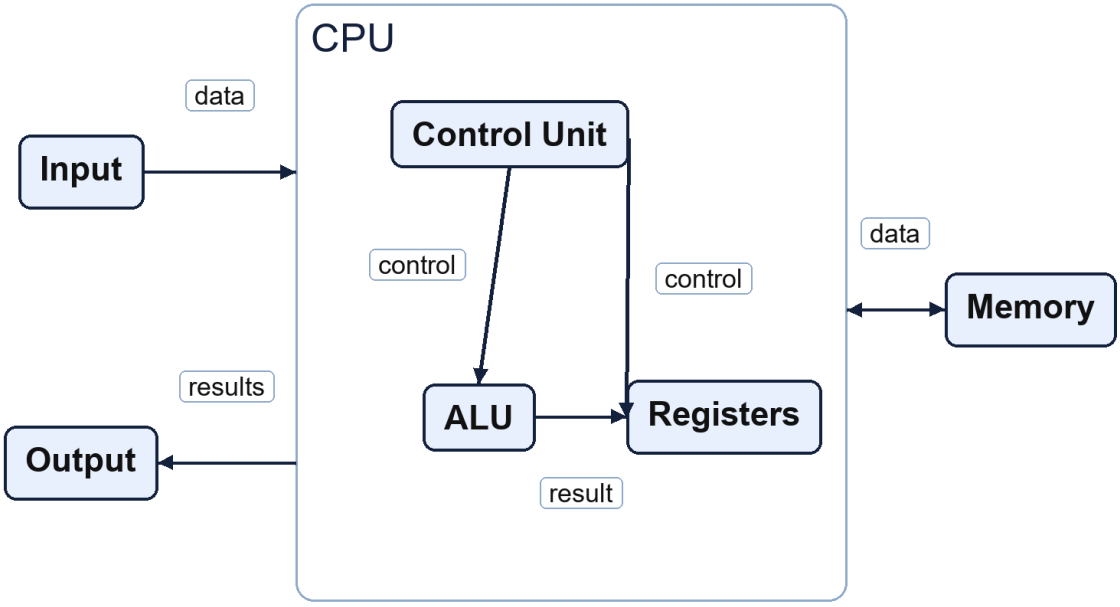


Figure 1: Stored-program computer organization.

(a) Name the computer organization shown in the figure and state its two key principles. (2 marks)

(b) Explain the function of the control unit and the arithmetic logic unit in this organization. (3 marks)

QUESTION A2 (5 marks)

The program counter and instruction register are important CPU registers.

(a) Differentiate the program counter and the instruction register. (2 marks)

(b) Describe the fetch-decode-execute cycle. (3 marks)

QUESTION A3 (5 marks)

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Answer the following data-representation questions. Show your complete workflow.

(a) Represent -85 in 8-bit sign-magnitude form. (2 marks)

(b) Represent -85 in 8-bit two's complement form. (3 marks)

QUESTION A4 **(5 marks)**

Answer the following data-representation questions. Show your complete workflow.

(a) Convert -156.15625 to IEEE 754 single-precision format. Give the result in hexadecimal and show the sign, exponent and mantissa fields. (5 marks)

QUESTION A5 **(10 marks)**

Answer the following data-representation questions. Show your complete workflow.

(a) Calculate -88 - (5) using 8-bit sign-magnitude. Show the complete workflow and the result in byte form. (5 marks)

(b) Calculate -88 - (5) using 8-bit two's complement. Show the complete workflow and the result in byte form. (5 marks)

QUESTION A6 **(10 marks)**

Study Figure 2 and answer the following questions.

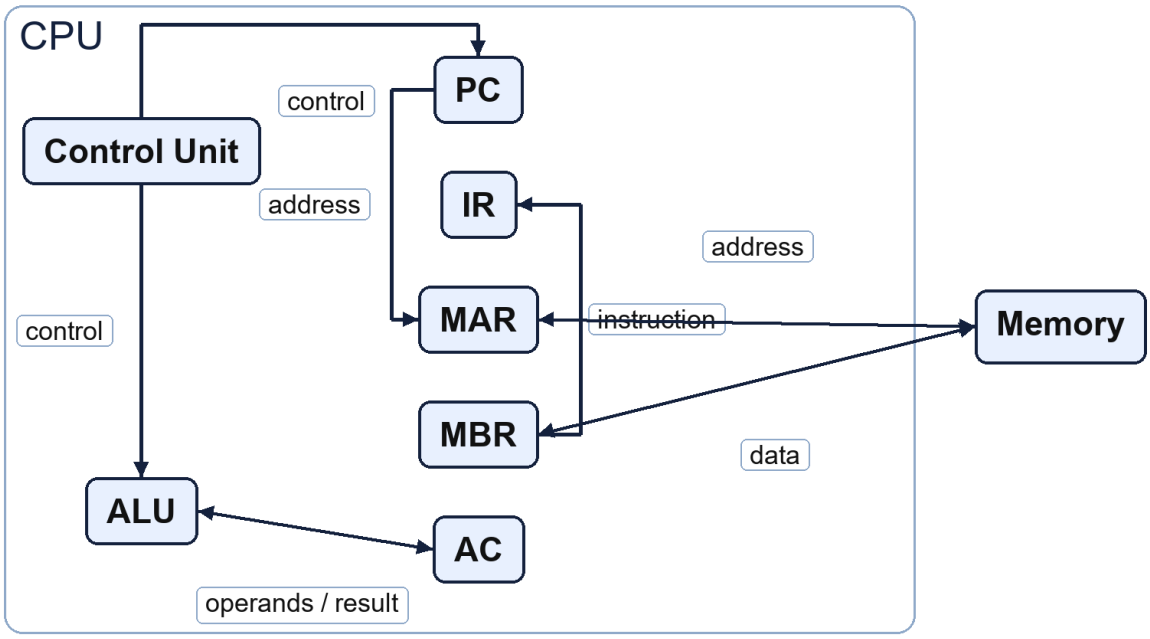


Figure 2: CPU, memory and system bus organization.

(a) Identify the register that holds the address of the next instruction and the register that holds the instruction currently being decoded. (2 marks)

(b) Explain the roles of MAR and MBR during a memory read. (3 marks)

(c) Describe the fetch-decode-execute cycle using the components shown in the figure. (5 marks)

QUESTION A7 **(10 marks)**

Buses connect the CPU, memory and I/O devices.

(a) Define a bus. (2 marks)

(b) Compare synchronous and asynchronous bus timing. (4 marks)

(c) Explain bus arbitration and why it is needed. (4 marks)

QUESTION A8 **(10 marks)**

Study Figure 3 and answer the following questions.

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Tag (4 bits)	Line (3 bits)	Offset (9 bits)

Cache state

Line	Valid	Tag	Data
0	0	----	----
1	0	----	----
2	0	----	----
3	1	0010	0xA5
4	0	----	----
5	1	0001	0x3C
6	0	----	----
7	0	----	----

Addresses
0x1A2B
0x2A2B

Figure 3: Direct-mapped cache organization and address fields.

(a) The cache uses a 16-bit address split into a 4-bit tag, a 3-bit line number and a 9-bit offset. For address 0x1A2B, give the tag, line number and offset in binary. (2 marks)

(b) Using the cache state shown in the figure, determine whether each address is a hit or a miss: (i) 0x1A2B, (ii) 0x2A2B. Explain your reasoning. (4 marks)

(c) Explain why the cache compares tags and how temporal and spatial locality improve the hit rate. (4 marks)

QUESTION A9 **(10 marks)**

The CPU can use programmed I/O, interrupt-driven I/O or DMA.

(a) Describe programmed I/O. (2 marks)

(b) Describe interrupt-driven I/O. (3 marks)

(c) Explain how DMA improves I/O performance and state ONE advantage over interrupt-driven I/O. (5 marks)

QUESTION A10 **(10 marks)**

Multicore processors are common in modern computers.

(a) Define multicore processing. (2 marks)

(b) Explain TWO benefits of multicore processing. (4 marks)

(c) Describe NUMA and ONE challenge in a distributed-memory system. (4 marks)

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Write appropriate programs using zero-, one- and three-address instructions to compute Z = (A + B) / (C - D). Registers A, B, C, D and E must not be modified; X, Y and Z are temporary registers; the final answer must be stored in Z. Available instructions: zero-address PUSH, POP, ADD, SUB, MUL, DIV; one-address LOAD, STORE, ADD, SUB, MUL, DIV; three-address ADD, SUB, MUL, DIV.

(a) Write a zero-address program for the expression. (6 marks)

(b) Write a one-address program for the expression. (6 marks)

(c) Write a three-address program for the expression. (8 marks)